Boosting Algorithm Assignment

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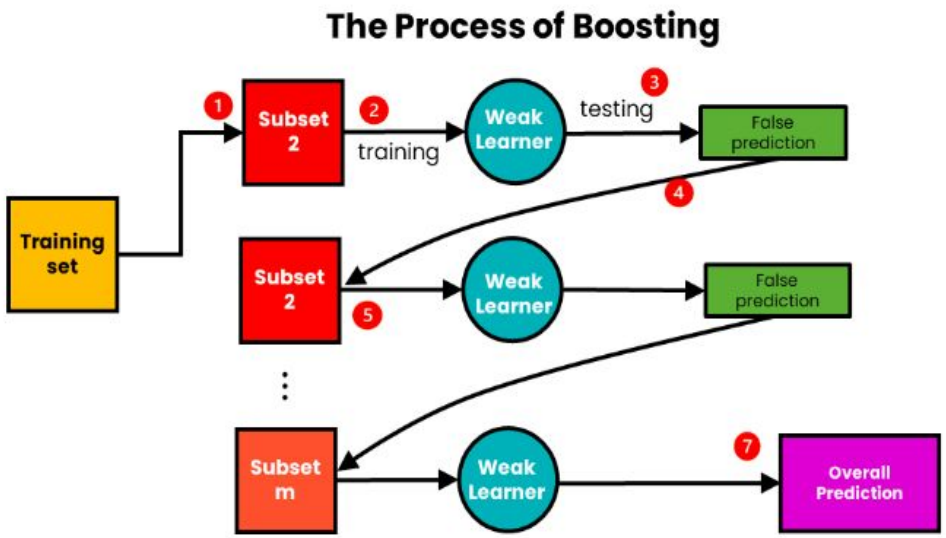
What is Boosting Algorithm?

Boosting is an Ensemble Learning method that converts a team of "weak learners" into a single "strong learner."

Unlike other methods that train models independently, boosting works sequentially. Each new model focuses specifically on the mistakes (residuals) made by the previous ones.

This can be used when you have a complex dataset where a single model isn't accurate enough, and you want to push for the highest possible precision.

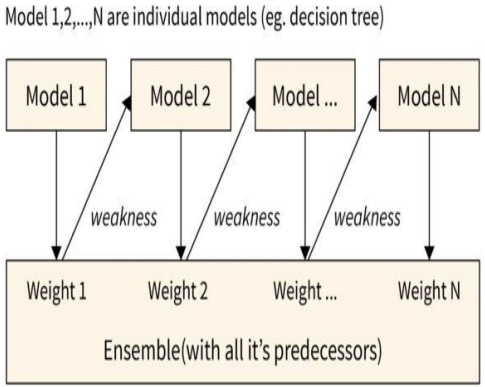


Adaptive Boosting

AdaBoost (Adaptive Boosting) assigns equal weights to all training samples initially and iteratively adjusts these weights by focusing more on misclassified data points for the next model. It effectively reduces bias and variance making it useful for classification tasks but it can be sensitive to noisy data and outliers.

The pioneer of boosting. It works by reweighting data. If a data point is misclassified, its "weight" is increased so the next model pays more attention to it.

Example - Face Detection: Classifying images as "Face" vs. "No Face."

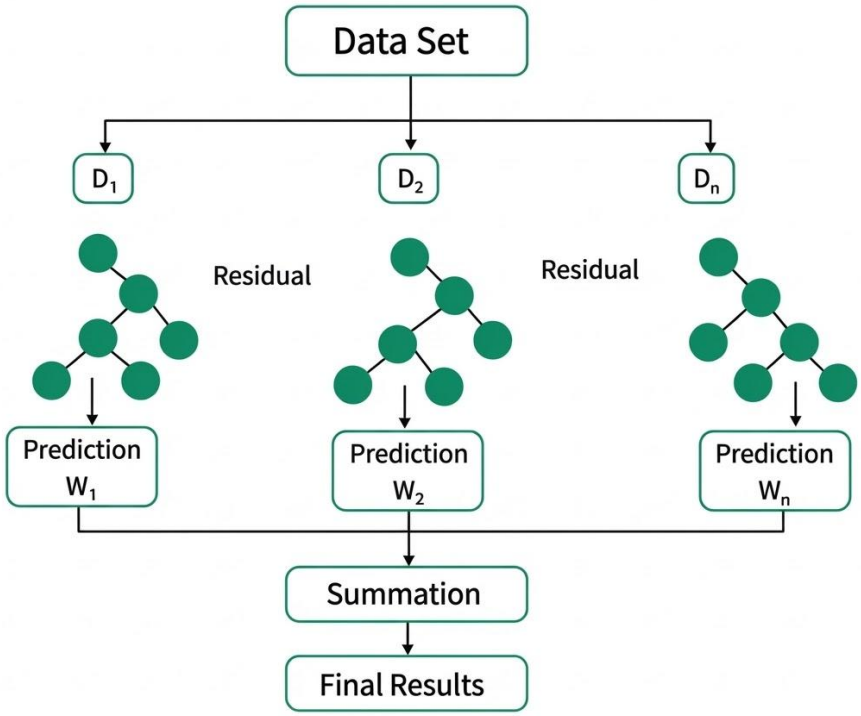


Extreme Gradient Boosting

An advanced version of Gradient Boosting built for speed and performance. It uses "Regularization" to prevent the model from becoming too complex and memorizing the data (overfitting).

It builds trees level-wise (growing horizontally across each level) and uses system-level optimizations like parallel processing.

Example - Predicting loan defaults or insurance claims where accuracy is critical

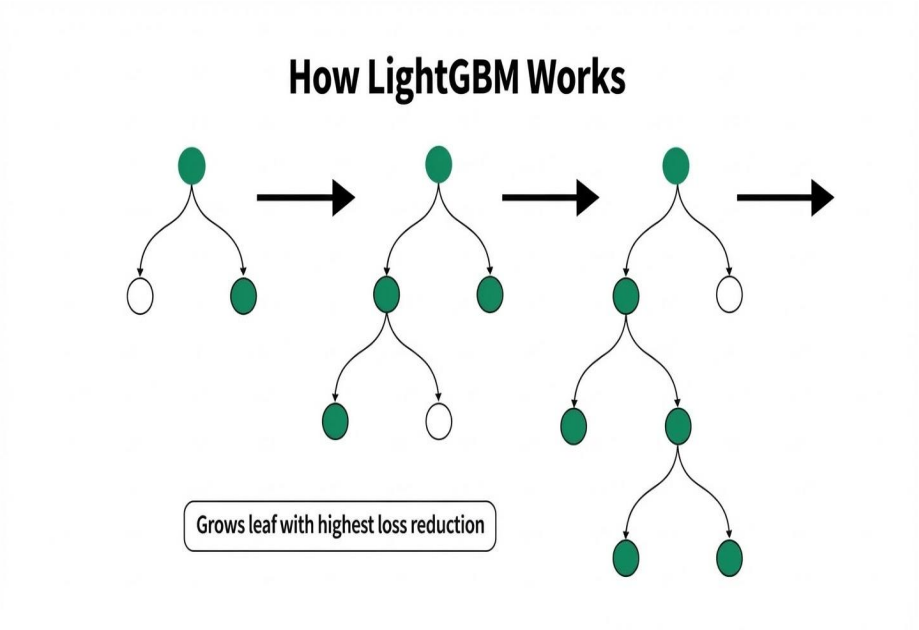


Light Gradient Boosting Machine

It grows trees leaf-wise (vertically). It picks the single most "broken" leaf and splits it further, rather than growing the whole level. This makes it much faster but requires careful tuning.

It is designed to handle massive datasets with lightning speed and low memory usage.

Example - Recommendation systems or search engine result ranking



Use Cases and Comparative Analysis

Algorithm	Strategy	Tree Growth	Best For
AdaBoost	Weighting Samples	Sequential Stumps	Simple binary classification
XGBoost	Gradient Descent	Level-wise	Medium data, high accuracy
LightGBM	Gradient Descent	Leaf Wise	Large-scale data, high speed